ECN 594: Identification and Instrumental Variables

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Plan for today

1. **Recap: Logit, Berry inversion, and identification**
2. Direction of bias
3. Uber example
4. Instrumental variables
5. Elasticities and limitations
6. The demographic interaction model
7. Why demographics help

Recap: Logit, Berry inversion, and identification

- Mean utility: $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} \quad (\alpha < 0)$
- **Berry inversion:**

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$$

- **The identification problem:**
  - Firms observe ξ_{jt} when setting prices
  - $\Rightarrow \text{Cov}(p_{jt}, \xi_{jt}) > 0 \Rightarrow$ OLS is biased
  - Need supply shifters (IVs) to identify demand
- **Today:** Direction of bias, IV strategies, and limitations

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Worked example: Bias direction

- **Question:** You estimate a logit demand model using OLS and get $\hat{\alpha} = -0.3$. A colleague says the true α is likely -0.5.
- Is this consistent with endogeneity bias? Why?

Worked example: Bias direction

- **Question:** You estimate a logit demand model using OLS and get $\hat{\alpha} = -0.3$. A colleague says the true α is likely -0.5.
- Is this consistent with endogeneity bias? Why?
- **Answer:** Yes!
  - OLS overstates how much consumers like expensive products
  - So OLS finds a smaller (less negative) price coefficient
  - $-0.3 > -0.5$, so this is exactly what we'd expect

Direction of bias

- If high-quality products have high prices...
- OLS sees: high price, but demand still high (because of ξ)
- OLS concludes: price doesn't hurt demand much
- **Result:** $\hat{\alpha}$ biased toward zero (less negative than truth)

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Case study: Uber surge pricing

- Uber's surge pricing: prices rise when demand is high
- **Why is surge pricing endogenous?**

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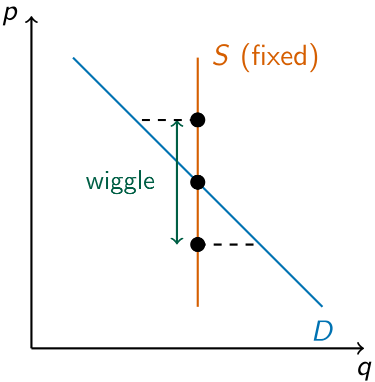
- Uber's surge pricing: prices rise when demand is high
- **Why is surge pricing endogenous?**
- Surge responds to BOTH:
    - **Supply shocks:** Fewer drivers available $\rightarrow$ surge up
    - **Demand shocks:** Concert ends, rain starts $\rightarrow$ surge up
- Can't just regress rides on surge multiplier!
- Same problem as before: observing equilibrium prices

The solution: Uber's price "wiggle" experiment

- Uber's insight: **randomize** prices!
- **Cohen et al. (2016):** Uber experimentally varied surge multipliers
- Same time, same location $\rightarrow$ different riders see different prices
- Key: Some riders randomly see 1.9x, others see 2.1x
- Demand conditions are identical, only price differs

Tracing out the demand curve



- Randomization holds supply fixed
- Price “wiggles” up/down randomly
- Traces out points ON the demand curve
- This identifies demand!

Uber experiment: Results

- Result: demand elasticity ≈ -0.5
- **Inelastic!** A 10% price increase $\rightarrow$ only 5% fewer rides
- Why so inelastic?
  - When you need a ride, you *need* a ride
  - Limited substitutes late at night
- **Key lesson:** Experiments give clean identification
- But most industries can't randomize prices...

Why experiments are powerful

- No confounding: price variation is independent of demand shocks
- Clean identification of the demand curve
- **But:**
  - Tech companies can run experiments
  - Traditional industries can't randomize prices
  - Most IO settings require **instrumental variables**

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Instrumental variables: the solution

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- Need variables z that are:
  1. **Relevant:** Correlated with price ($\text{Cov}(z, p) \neq 0$)
  2. **Exogenous:** Uncorrelated with ξ ($\text{Cov}(z, \xi) = 0$)
- These are cost shifters or other supply-side variables

Common IVs in demand estimation

Three main approaches used in the literature:

1. **Cost shifters:** Input prices, exchange rates
2. **Hausman IVs:** Prices in other markets
3. **BLP IVs:** Characteristics of competing products

Let's examine each in detail...

IV #1: Cost shifters

- **Idea:** Use variables that shift marginal costs
  - Input prices (steel, labor, energy)
  - Exchange rates (for imported goods)
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- **Exogeneity:** Cost shocks don't directly affect consumer utility
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- **Problem:** Cost data rarely observed!
  - Firms don't disclose detailed cost information
  - Even aggregate input prices may be hard to match to products

IV #2: Hausman instruments

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- **Idea:** Use prices of the same product in *other markets*
  - Coca-Cola price in Chicago as IV for Coca-Cola price in Phoenix
- **Why it works:**
  - Same product has similar costs across markets
  - Cost shocks (ingredients, shipping) affect prices everywhere
- **Key assumption:** Demand shocks ξ_{jt} are independent across markets
  - Chicago's unobserved demand factors don't affect Phoenix

Worked example: When do Hausman IVs fail?

- **Question:** You want to use Pepsi prices in Denver as an IV for Pepsi prices in Phoenix. When might this fail?

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- **Question:** You want to use Pepsi prices in Denver as an IV for Pepsi prices in Phoenix. When might this fail?
- **Answer:**
- Hausman IVs assume ξ_{jt} is independent across markets
- **This fails when:**
  - National advertising campaigns affect all markets
  - Nationwide product reformulation ("New Coke")
  - Coordinated promotions or seasonal demand patterns
- **Relevance is OK:** Cost shocks do affect prices everywhere
- **Exogeneity is the problem:** Demand shocks may also be correlated!

IV #3: BLP instruments

- **Idea:** Use characteristics of *other products* in the market

- **Key assumption:**

$$E(\xi_{jt}|x_t) = 0$$

"Observed characteristics are mean independent of unobserved characteristics"

- **Common BLP instruments:**

  1. Own product characteristics (not price!)

  2. Sum of characteristics of *own-firm* products

  3. Sum of characteristics of *competitor* products

- **Intuition:** More/closer competitors $\rightarrow$ more competition $\rightarrow$ lower prices

BLP instruments: Example

- Consider estimating demand for the Honda Civic
- **BLP instruments could include:**
  - Civic's own characteristics: horsepower, MPG, size
  - Sum of Honda's other cars' characteristics (Accord, CR-V)
  - Sum of competitors' characteristics (Corolla, Mazda3, etc.)

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  - Sum of Honda's other cars' characteristics (Accord, CR-V)
  - Sum of competitors' characteristics (Corolla, Mazda3, etc.)
- **Why exogenous?**
  - Toyota's decision about Corolla's horsepower doesn't affect Civic's unobserved quality ξ
- **Why relevant?**
  - If Toyota improves Corolla, Honda faces more competition
  - This pressures Honda to lower Civic's price

IV #4: Fan/Waldfogel instruments

- **Idea:** Use demographics in *other markets* where the product is sold
- Example: Fan (2013) - Newspaper demand
  - Newspapers sold in multiple counties
  - Use demographics in other counties as IVs for price

IV #4: Fan/Waldfogel instruments

- **Idea:** Use demographics in *other markets* where the product is sold
- Example: Fan (2013) - Newspaper demand
  - Newspapers sold in multiple counties
  - Use demographics in other counties as IVs for price
- **Why it works:**
  - Product characteristics shaped by demographics in ALL markets where sold
  - Demographics elsewhere affect costs/design but not local demand
- **Validity:** ξ_{jt} not correlated across markets (same as Hausman)
- **Concern:** Which markets a firm enters may be endogenous

Worked example: Evaluating an IV

- **Question:** A researcher proposes using gasoline prices as an IV for car prices. Is this valid?

Worked example: Evaluating an IV

- **Question:** A researcher proposes using gasoline prices as an IV for car prices. Is this valid?
- **Relevance:** Higher gas prices $\rightarrow$ higher operating costs $\rightarrow$ might affect car prices? Maybe weakly.
- **Exogeneity:** Do gas prices affect car quality ξ_{jt}?
  - Gas prices affect *demand* for fuel-efficient cars
  - This might shift which cars look "good" to consumers
  - Potentially problematic!
- **Overall:** Probably not a great IV

Summary: IV conditions

- For z to be a valid IV:
  1. **Relevant:** z must predict prices
     - Can test this! Run first-stage regression
  2. **Exogenous:** z must not affect demand directly
     - Cannot test this directly (requires economic reasoning)
- This is the standard IV framework from econometrics
- Applied to demand estimation: use supply-side variation

What makes a good IV? Summary

Instrument	Relevance	Exogeneity
Cost shifters (input prices)	Strong	Usually OK
Hausman IVs (prices elsewhere)	Strong	Debatable
BLP IVs (competitor chars)	Medium	Usually OK
Fan/Waldfogel (demographics elsewhere)	Medium	Debatable
Experiments (Uber wiggle)	Perfect	Perfect

- No perfect instrument in observational data
- **Best practice:** Use multiple IVs, check robustness
- **Experiments** are gold standard but often infeasible

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From estimation to elasticities

- We've seen how to estimate demand with logit
- **Why do we care?**
- Demand estimates let us compute **elasticities**
  - How sensitive are consumers to price changes?
  - How do consumers substitute between products?
- These are crucial for policy: mergers, pricing, welfare
- **But:** the logit model has important limitations...

Logit elasticities

- Given shares $s_{jt} = \frac{\exp(\delta_{jt})}{1+\sum_k \exp(\delta_{kt})}$

- We can derive price elasticities:

$$\eta_{jj} = \frac{\partial s_{jt}}{\partial p_{jt}} \frac{p_{jt}}{s_{jt}} = \alpha p_{jt}(1 - s_{jt}) \quad \text{(own-price)}$$

$$\eta_{jk} = \frac{\partial s_{jt}}{\partial p_{kt}} \frac{p_{kt}}{s_{jt}} = -\alpha p_{kt} s_{kt} \quad \text{(cross-price)}$$

- Note: $\alpha < 0$, so own-price elasticity is **negative** (as expected)

Limitation #1: Own-price elasticity

- Own-price elasticity: $\eta_{jj} = \alpha p_{jt}(1 - s_{jt})$

- With many products: $(1 - s_{jt}) \approx 1$

- So: $\eta_{jj} \approx \alpha p_{jt}$

- **Implication:** Own-price elasticity is proportional to price!

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- **Implication:** Own-price elasticity is proportional to price!

- Lower-priced products have more *inelastic* demand

- Higher-priced products have more *elastic* demand

- **Question:** Is this realistic?
    - Generic cereal (low price) should be more elastic than organic cereal?
    - Budget airline (low price) more inelastic than first class?

Limitation #2: Unrealistic substitution patterns

- Cross-price elasticity: $\eta_{jk} = -\alpha p_{kt} s_{kt}$
- This doesn't depend on product j at all!
- **Implication:** All products have the *same* cross-elasticity with product k

Limitation #2: Unrealistic substitution patterns

- Cross-price elasticity: $\eta_{jk} = -\alpha p_{kt} s_{kt}$
- This doesn't depend on product j at all!
- **Implication:** All products have the *same* cross-elasticity with product k
- Example: BMW raises its price
  - Logit says: same fraction go to Mercedes as to Honda Civic!
  - Reality: BMW consumers mostly switch to Mercedes, Audi
- Substitution depends on **market share**, not product similarity

Diversion ratios in logit

- When product k's price rises, where do consumers go?
- **Diversion ratio** from k to j:

$$D_{k \to j} = \frac{\partial s_{jt} / \partial p_{kt}}{\partial s_{kt} / \partial p_{kt}} = \frac{s_{jt}}{1 - s_{kt}}$$

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- Substitution is proportional to **market share**, not similarity!
- **Why this matters:**
  - Merger analysis: which products are close substitutes?
  - Product positioning: who are my real competitors?
  - Logit can't answer these questions well

Why logit has unrealistic substitution

- The problem comes from the **logit error structure**
- Key assumption: ε_{ijt} are *independent* across products
- This means: your taste shock for BMW is unrelated to your taste shock for Mercedes
- The model doesn't "know" that BMW and Mercedes are similar

Why logit has unrealistic substitution

- The problem comes from the **logit error structure**
- Key assumption: ε_{ijt} are *independent* across products
- This means: your taste shock for BMW is unrelated to your taste shock for Mercedes
- The model doesn't "know" that BMW and Mercedes are similar
- **Solution:** Allow for consumer heterogeneity
  - High-income consumers prefer both BMW and Mercedes
  - When BMW's price rises, these consumers switch to Mercedes

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The demographic interaction model

- We estimate a version of **Berry, Levinsohn & Pakes (1995)** — "BLP"
- Allow preferences to vary across consumers:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

- β_i, α_i: consumer i's taste for characteristics and price
- **Key question:** What drives heterogeneity in β_i and α_i?
- **Our approach:** Observed demographics

$$\beta_{ik} = \beta_k + \sum_{\ell=1}^{L} \pi_{k\ell} D_{i\ell}, \qquad \alpha_i = \alpha + \sum_{\ell=1}^{L} \pi_{\alpha\ell} D_{i\ell}$$

- $D_{i\ell}$: demographic ℓ for consumer i (income, age, family size, etc.)

Rewriting in terms of mean utility

- Substituting the expressions for β_{ik} and α_i:

$$u_{ijt} = \underbrace{x_{jt}\beta + \alpha p_{jt} + \xi_{jt}}_{\delta_{jt} \text{ (mean utility)}} + \underbrace{\sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k} + \sum_{\ell} \pi_{\alpha\ell} D_{i\ell} p_{jt}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- **Mean utility** δ_{jt}: part common to all consumers
  - Contains "**linear parameters**" (β, α) — enter linearly in δ
- **Individual deviation** μ_{ijt}: how consumer i differs from average
  - Contains "**nonlinear parameters**" $(\pi_{k\ell})$ — enter nonlinearly via shares
- Note: $\alpha < 0$ (higher price $\rightarrow$ lower utility)

From utility to demand

- **Individual choice probability** (with T1EV errors):

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt} + \mu_{ikt})}$$

- **Market share** (aggregate over consumers):

$$s_{jt} = \frac{1}{I} \sum_{i=1}^{I} s_{ijt}$$

- This is **demand**: the fraction of consumers who buy product j

- Key: shares depend on mean utilities δ and demographics D_i

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Elasticities with demographic interactions

- **Elasticities with heterogeneous consumers:**

$$\eta_{jkt} = \begin{cases} \frac{p_{jt}}{s_{jt}} \sum_i \alpha_i s_{ijt}(1 - s_{ijt}) & \text{if } j = k \\ -\frac{p_{kt}}{s_{jt}} \sum_i \alpha_i s_{ijt} s_{ikt} & \text{otherwise} \end{cases}$$

- s_{ijt}: probability that consumer i purchases product j
- **Key insight:** Cross-elasticity now depends on $\sum_i s_{ijt} s_{ikt}$
  - Products that attract *similar consumers* have higher cross-elasticity
  - BMW and Mercedes attract similar (high-income) consumers $\Rightarrow$ closer substitutes
- Correlation in μ_{ijt} and μ_{ikt} drives realistic substitution patterns

Limitations: Demographics vs Mixed Logit

- **Demographics model** (what we use):
  - Heterogeneity only from observed characteristics
  - Substitution still restrictive within demographic groups
  - Simple and tractable
- **Mixed logit / random coefficients** (advanced):
  - Heterogeneity from unobserved factors too
  - Fully flexible substitution patterns
  - Computationally intensive (requires simulation)
- For this course: demographics are sufficient
- Know that mixed logit exists for research

Key Points

1. **Price is endogenous**: Firms observe ξ_{jt} when pricing $\rightarrow$ OLS biased toward zero
2. **Solution**: Instrumental variables (cost shifters, BLP IVs, Hausman IVs)
3. **Logit limitation**: Cross-elasticity doesn't depend on j $\Rightarrow$ unrealistic substitution
4. **Demographic model**: $u_{ijt} = \delta_{jt} + \sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k} + \varepsilon_{ijt}$
5. Demographics $\Rightarrow$ heterogeneous preferences $\Rightarrow$ realistic substitution